

Week 3: Games with Simultaneous Moves

Nash Equilibrium, Best Responses & Strategic Market Stability

DOT 3090: Strategic Thinking: Game Theory for Business Strategy

Course Instructor: **Prof. Chiu Yu Ko**

Institution: Department of Decision Sciences and Managerial Economics, CUHK

Term & Academic Year Semester 2026-27 Term 1 | MBA & Executive Master Curriculum

Central Principle:

"In simultaneous moves, neither firm observes the other's action before choosing.

Strategic stability requires every player to choose their best response to rivals' choices."

Opening Decision: Duopoly Capacity Choice

Motivating Case

The Situation: Airbus vs. Boeing Wide-Body Production Planning

Both aircraft manufacturers must finalize annual wide-body production schedules simultaneously. Neither firm can observe the rival's exact assembly rate before committing billions to supply chains.

		Player 2: Boeing	
		Mod Capacity	High Capacity
Player 1: Airbus	Mod Capacity	\$8B , \$8B	\$2B , \$10B
	High Capacity	\$10B , \$2B	\$4B , \$4B Nash Eq

Strategic Diagnosis:

- ¢ If Boeing chooses Moderate, Airbus earns \$10B by choosing High vs \$8B choosing Moderate.
- ¢ If Boeing chooses High, Airbus earns \$4B by choosing High vs \$2B choosing Moderate.
- ¢ **High Capacity is Airbus's strictly dominant strategy! Symmetric logic yields High for Boeing.**

Today's Learning Outcomes

Session Objectives

By the conclusion of this quantitative workshop, executives will be able to:

1

Formalize Best Response Functions:

Compute optimal strategic actions conditional on any rival choice.

2

Identify Nash Equilibria:

Detect strategically stable profiles using cell inspection and best response crossings.

3

Execute IESDS:

Systematically prune strictly dominated moves to simplify complex multidimensional games.

4

Diagnose the 4 Classic 2x2 Archetypes:

Prisoner's Dilemma, Coordination Game, Hawk-Dove, and Matching Pennies.

5

Compare Cournot vs. Bertrand Competition:

Contrast simultaneous quantity choice against cutthroat price competition.

Strategic Foundations: Simultaneous Moves Defined

What 'Simultaneous' Really Means:

- ¢ NOT that decisions occur on the exact same second.
- ¢ It means: NO player observes the other's choice before executing their own strategic action.
- ¢ **Information sets are imperfect and concurrent.**

Corporate Examples in the Wild:

- ¢ Sealed-bid public procurement and spectrum tenders.
- ¢ Annual Black Friday product discounting & promotions.
- ¢ Fab node capacity investments in semiconductor foundry.
- ¢ Next-gen video console launch prices (Sony vs MSFT).

The Mental Shift from Sequential to Simultaneous Analysis:

1. In Sequential Games (Week 2): You look ahead and see what rival DID or WILL do at subgames.
2. In Simultaneous Games (Week 3): You must simultaneously form a rational conjecture about rival moves.
3. Equilibrium requires beliefs to be mutually consistent: each firm accurately anticipates the other.
4. **Neither firm has regret post-revelation: no incentive to deviate unilaterally.**

The Normal Form (Payoff Matrix) Representation

Analytical Tool

Standard 3-Tuple Specification of a Simultaneous Game: $G = \langle N, \{S_i\}, \{u_i\} \rangle$

Key Matrix Components:

- 1. $N = \{1, 2, \dots, n\}$ Players
- 2. S_i : Strategy set of player i
- 3. $u_i(s_1, s_2)$: Payoff function

Convention:

- ⌘ Row Player payoff = 1st number
- ⌘ Col Player payoff = 2nd number

Player 1: Firm 1
Action L

Action R

Player 2: Firm 2
Action U Action D

$u_1(L,U)u_2(L,U)$	$u_1(L,D)u_2(L,D)$
$u_1(R,U)u_2(R,U)$	$u_1(R,D)u_2(R,D)$

Managerial Diagnostic Checklist:

- ⌘ Check completeness: Are all realistic executive actions enumerated in strategy sets S_i ?
- ⌘ Check payoff symmetry vs asymmetry: Do firms face identical cost curves or legacy handicaps?

Best Responses: The Core Mathematical Principle

Equilibrium Foundation

Mathematical Definition of Best Response Function $BR_i(s_{-i})$:

A strategy s_i^* is player i 's Best Response to rival profile s_{-i} if and only if:

$$u_i(s_i^*, s_{-i}) \geq u_i(s_i, s_{-i}) \quad \text{for all other available strategies } s_i \text{ in } S_i$$

It answers the fundamental executive question: 'Given what rival chooses, what maximizes my return?'

Strict vs. Weak Dominance:

☛ Strictly Dominant Strategy:

BR to EVERY possible rival strategy.

☛ Strictly Dominated Strategy:

Another action yields strictly higher payoff regardless of what rival does.

Rule: Rational players NEVER play dominated moves!

The Underline Algorithm (Step-by-Step):

1. Hold Player 2's action fixed in Col 1.

Underline Player 1's highest payoff in that column.

2. Repeat for all remaining columns.

3. Hold Player 1's action fixed in Row 1.

Underline Player 2's highest payoff in that row.

4. Any cell with BOTH payoffs underlined is a Nash Eq!

Iterated Elimination of Strictly Dominated Strategies (IESDS)

Pruning Non-Rational Strategic Noise from High-Dimensional Decisions

The 3 Axioms of IESDS in Business Analysis:

Axiom 1: A rational executive will NEVER choose a strictly dominated action under any scenario.

Axiom 2: Knowing rival is rational, you anticipate they will NEVER play their dominated actions either.

Axiom 3: Pruning dominated rows/cols may render previously undominated actions dominated in round 2.

Guaranteed Property: IESDS NEVER eliminates any Nash Equilibrium! Order of deletion does not matter.

Managerial Power of Dominance Solvability:

☞ When IESDS reduces the game to a single unique cell, the game is called Dominance Solvable.

☞ You do NOT need mutual trust, coordination, or communication to predict the outcome.

☞ Common knowledge of rationality alone guarantees that players will converge to this equilibrium.

☞ **Caveat: If rival is non-rational or faces hidden private payoffs, IESDS predictions fail!**

What is a Nash Equilibrium?

Core Theory

Formal Definition (John Forbes Nash Jr., 1950):

A strategy profile $(s_1^*, s_2^*, \dots, s_n^*)$ is a Nash Equilibrium if for every player i :

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \text{for all possible actions } s_i$$

Plain English: No player can unilaterally improve their payoff by changing their own action alone.

Why Nash Equilibrium Matters in Business:

- ¢ Strategic Stability: It is self-enforcing.
- ¢ No player experiences ex-post regret.
- ¢ If a non-equilibrium state is proposed, at least one firm has a private incentive to cheat or renege secretly.
- ¢ **Therefore, cartels without enforcement collapse.**

Common Executive Fallacies to Avoid:

- Fallacy 1: 'Nash Equilibrium maximizes total profit.'**
False! Prisoner's dilemma equilibrium is Pareto-inferior.
- Fallacy 2: 'Every game has exactly one Nash Eq.'**
False! May have 0 pure eq (Matching Pennies) or 2+ (Coordination).
- Fallacy 3: 'Nash Equilibrium requires cooperation.'**
False! It assumes ruthless non-cooperative self-interest.

Finding Pure Strategy Nash Equilibria

Technique & Demonstration

Step-by-Step Demonstration of the Underline Best-Response Algorithm:

		Player 2: Firm 2	
		Strategy X	Strategy Y
Player 1: Firm 1	Strategy A	5 , 4	1 , 6
	Strategy B	7 , 2	3 , 3 Nash Eq

- Step-by-Step Verification:**
- ¢ If Firm 2 plays X: Firm 1 compares 5 (A) vs 7 (B) -> Firm 1 chooses B. Underline 7.
 - ¢ If Firm 2 plays Y: Firm 1 compares 1 (A) vs 3 (B) -> Firm 1 chooses B. Underline 3. (B is dominant for P1!)
 - ¢ If Firm 1 plays B: Firm 2 compares 2 (X) vs 3 (Y) -> Firm 2 chooses Y. Underline 3.

Archetype 1: The Prisoner’s Dilemma

Classic 2x2 Games

Dominant Defection & Joint Profit Destruction in Price Wars

		Player 2: Firm 2	
		Cooperate (High P)	Defect (Discount)
Player 1: Firm 1	Cooperate (High P)	\$10M, \$10M	\$2M , \$14M
	Defect (Discount)	\$14M, \$2M	\$4M , \$4M Nash Eq

Strategic Diagnosis & Executive Lessons:

- ¢ Structure: Each player has a strictly dominant strategy to Defect, regardless of rival move.
- ¢ Paradox: Unique Nash Eq [Defect, Defect] (\$4M, \$4M) is Pareto-dominated by [Coop, Coop] (\$10M, \$10M).
- ¢ Business Manifestations: Food-delivery coupons, airline seat wars, aggressive digital ad spending.

Archetype 2: Coordination Games & Standard Wars

Classic 2x2 Games

Multiple Equilibria, Network Externalities, and Focal Points

		Player 2: Microsoft	
		Standard Alpha	Standard Beta
Player 1: Sony	Standard Alpha	\$12B , \$8B Nash Eq	\$0B , \$0B
	Standard Beta	\$0B , \$0B	\$8B , \$12B Nash Eq

Strategic Diagnosis & Executive Lessons:

- ¢ Structure: TWO pure Nash Equilibria: (Alpha, Alpha) and (Beta, Beta). Miscoordination yields zero!
- ¢ Key Problem: How do players coordinate without formal merger? Requires Focal Points (Schelling Points).
- ¢ Real Cases: EV charging standards (Tesla NACS vs CCS), Blu-ray vs HD-DVD, 5G wireless protocols.

Archetype 3: Hawk-Dove / Chicken & Brinkmanship

Classic 2x2 Games

Anti-Coordination, Mutually Destructive Collisions, and Preemption

		Player 2: Firm 2	
		Accommodate (Dove)	Aggressive (Hawk)
Player 1: Firm 1	Accommodate (Dove)	\$4M , \$4M	\$1M , \$8M Nash Eq
	Aggressive (Hawk)	\$8M , \$1M Nash Eq	-\$10M , -\$10M

Strategic Diagnosis & Executive Lessons:

- ¢ Structure: Two asymmetric equilibria where ONE firm acts aggressive and the other backs down.
- ¢ Worst Outcome: Mutual aggression triggers mutual ruin (-\$10M, -\$10M).
- ¢ Winning Strategy: Irreversible commitment to Hawk (e.g. public contractual penalties or burned bridges).

Archetype 4: Matching Pennies & Mixed Strategies

Classic 2x2 Games

Zero-Sum Games, Cyclical Incentives, and Unpredictability

		Player 2: Target Firm	
		Comply / Heads	Cheat / Tails
Player 1: Inspector Audit / Heads		+1 , -1	-1 , +1
	Pass / Tails	-1 , +1	+1 , -1

- Strategic Diagnosis & Executive Lessons:
- ¢ Structure: Pure conflict of interest. NO pure strategy Nash equilibrium exists!
 - ¢ Cyclical Trap: If Target complies, Inspector passes; if Inspector passes, Target cheats; if Target cheats...
 - ¢ Solution Preview (Week 4): Mixed strategies (deliberate randomization to stay unpredictable).

Cournot (Quantity) Competition in Oligopoly

Market Structure

The Cournot Model Architecture:

- ¢ Dual firms choose quantities q_1 , q_2 simultaneously.
- ¢ Market clears at price $P(Q) = a - b(q_1 + q_2)$.
- ¢ Best Response Function $q_1^*(q_2)$ is downward sloping:
 $q_1^*(q_2) = (a - c - b \cdot q_2) / (2b)$

Strategic Equilibrium Properties:

- ¢ Quantities are strategic substitutes: Rival expands $\rightarrow$ you contract.
- ¢ Symmetric Cournot equilibrium: $q_1^* = q_2^* = (a - c) / (3b)$.
- ¢ Market Price sits strictly between Monopoly and Perfect Competition:
 $P_{\text{monopoly}} > P_{\text{cournot}} > P_{\text{competitive}} = MC$

Where Cournot Best Describes Business Reality:

- ¢ Heavy industries requiring irreversible advance capacity investments:
 - Commercial aircraft assembly capacity (Boeing vs Airbus)
 - Semiconductor wafer fab allocation (TSMC vs Samsung vs Intel Foundry)
 - Global crude oil production quotas (OPEC+ member production)
- ¢ **Takeaway: When production lead times are long, capacity choices determine final market power.**
Capacity preemption acts as an effective commitment device.

Bertrand (Price) Competition & The Bertrand Trap

Market Structure

The Bertrand Model (Price Competition):

- ¢ Dual firms choose prices p_1, p_2 simultaneously.
- ¢ Products are homogeneous; buyers buy lowest price.
- ¢ Undercutting incentive: shaving price by \$0.01 steals 100% of market demand.

The Bertrand Paradox:

- ¢ **Unique Nash Eq: $p_1^* = p_2^* = \text{Marginal Cost (c)}$.**
- ¢ **With ONLY TWO FIRMS, economic profits drop to ZERO!**
- ¢ Oligopoly collapses into perfect competition pricing.
- ¢ *Devastating for executive shareholder returns.*

How Strategic Executives Escape the Bertrand Trap:

1. Product Differentiation: Build brand moats, proprietary UI, ecosystem lock-in (Apple vs Android).
2. Capacity Constraints: Edgeworth modification - if rival cannot serve all demand, price stays above MC.
3. Customer Switching Costs: Contractual terms, enterprise integration API locks, data stickiness.
4. Tacit Collusion via Repeated Play: Escaping short-term price cuts to maintain long-term margins (Week 6).

Core Lesson: Never compete on identical price alone without structural protection.

Business Case 1: Airbus vs. Boeing

Case Applications

Wide-Body Aircraft Preemption & Duopoly Supply Discipline:

- ¢ The commercial aircraft sector is a duopoly protected by multi-billion dollar R&D barriers.
- ¢ When designing next-gen aircraft (A350 vs B787), both firms faced capacity ramp-up decisions.
- ¢ If both build aggressive assembly lines, market oversupply slashes lease rates and margins.

Game Theoretic Analysis:

1. Capacity commitment is irreversible: Tooling and supplier long-term contracts lock in output.
2. Cournot reaction functions apply: High capacity by Boeing depresses Airbus's residual marginal revenue.
3. Preemptive signaling: Announcing firm orders with major airlines (Emirates, United) shifts rival expectations.

Managerial Takeaway:

In capital-intensive industries, capacity preemption acts as an extensive-form bridge into a simultaneous **production game, forcing the competitor along their downward-sloping best response curve.**

Business Case 2: Sony PlayStation vs. Microsoft Xbox

Case Applications

Hardware Subsidization & Simultaneous Platform Entry Games:

- ¢ Console cycles (PS3 vs Xbox 360, PS4 vs Xbox One, PS5 vs Xbox Series X) feature intense simultaneous pricing.
- ¢ Two-sided market dynamics: Installing an active gamer base attracts third-party game publishers (EA, Ubisoft).
- ¢ Subsidizing hardware at a loss (selling at \$399 when BOM cost is \$450) is compensated by royalty cuts.

Game Theoretic Analysis:

- ¢ In 2013 (E3 Conference), Microsoft announced Xbox One at \$499 (bundled with Kinect sensor).
- ¢ Hours later, Sony announced PS4 at \$399, exploiting Microsoft's miscalculated cost structure.
- ¢ In 2020 (PS5 vs Series X), both firms waited until weeks before holiday launch to reveal price simultaneously.

Managerial Takeaway:

In two-sided platforms, hardware price is not an isolated margin driver, but the gateway to life-time **software royalties. The Nash Equilibrium entails pricing hardware close to marginal cost.**

Business Case 3: OPEC+ Crude Oil Quotas & Cheating

The Multilateral Cartel Prisoner's Dilemma:

- ☐ Global oil demand is inelastic in the short run; collective production cuts boost global barrel price.
- ☐ Cartel Agreement: Saudi Arabia, UAE, Russia agree to strict output quotas.
- ☐ Individual Cheating Incentive: Once price is high (\$85/bbl), each member earns millions by pumping +5%.

Game Theoretic Analysis:

- ☐ In a one-shot simultaneous game, Overproducing is each member's strictly dominant strategy.
- ☐ If all overproduce, global oil crashes to \$35/bbl, harming all producers.
- ☐ Sustaining Cartel Discipline requires satellite tanker tracking (monitoring) and credible punishment (Saudi flood).

Managerial Takeaway:

No collusive agreement can survive on goodwill alone. A Nash Equilibrium requires that defecting is immediately **detected and countered with retaliatory capacity**.

Executive Synthesis: 5 Strategic Rules of Thumb Executive Summary

Strategic Checklist for Decision-Makers in Simultaneous Competitive Arenas:

Rule 1

Never Play Dominated Moves:

Systematically audit executive options; delete actions that lose regardless of rival actions.

Rule 2

Anticipate Rival Rationality:

Do not assume your competitors will make obvious mistakes; model their best responses.

Rule 3

Seek Self-Enforcing Deals:

Any strategic partnership without aligned incentives will collapse into defection at equilibrium.

Rule 4

Choose the Dimension Wisely:

Compete on Capacity/Quantity (Cournot) when possible; avoid homogeneous Price wars (Bertrand).

Rule 5

Transform the Game:

If trapped in a destructive Nash Eq, change the payoffs: differentiate, merge, or alter timing.

Looking Ahead to Week 4: Mixed Strategies

Next Session Roadmap

Preview: Mixed Strategies & Randomization in Business Strategy

- ¢ What happens when a game has NO Pure Strategy Nash Equilibrium (like Matching Pennies)?
- ¢ How do retail platforms (Amazon, Target) use algorithmic price randomization to prevent competitor price-matching bots?
- ¢ **The Indifference Principle: How to calculate optimal mixed strategy probability distributions.**

Recommended Week 3 Post-Workshop Assignments:

1. Review the Full Lecture Recap in this portal (interactive payoff models & matrix inspector).
 2. Solve the Airbus vs. Boeing Case Study in the 'Case & Payoff Model' tab.
 3. Complete Week 3 Strategy Quiz (5 scenario-based executive questions).
 4. Experiment with custom 2x2 matrices in the embedded Strategy Simulator & GameTikzStudio.
- Faculty X Primer: <https://x.com/kochiuyu/status/2040209356831199262>*